

### 3 ENERGY BAND OF SEMICONDUCTOR

#### 3.1 Molecular orbitals and energies

To understand the energy band formation in solid, let us, first of all, consider the formation of energy levels when two hydrogen atoms are brought together to form a hydrogen molecule ( $H_2$ ). The  $H_2$  molecule are bonded by the covalent bonding.

- The electronic configuration of H atoms is

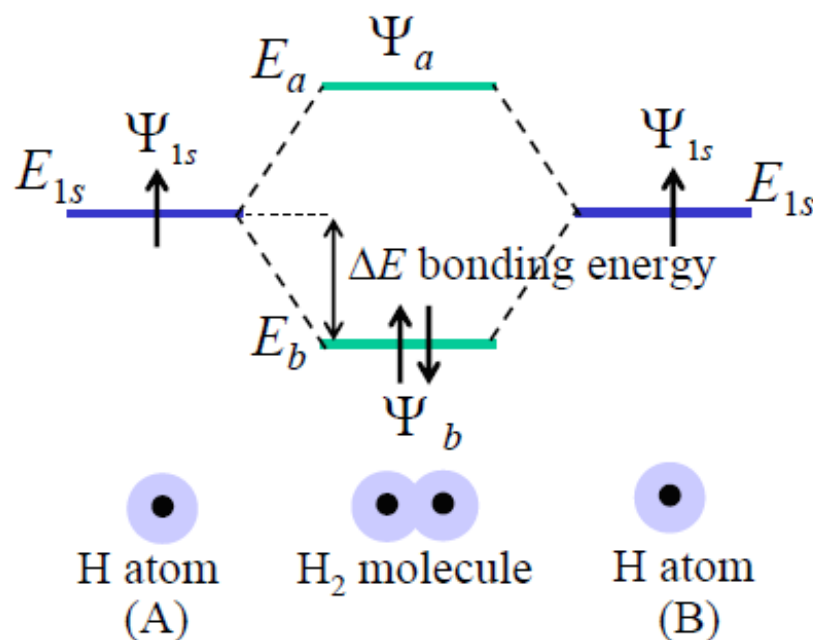
$$H: 1s^1.$$

- The wave function of the  $1s$  state:  $\Psi_{1s}$

$\Psi_{1s}$  describes a quantum state with  $n = 1$ ,  $l = 0$  and  $m_l = 0$

- The energy of the  $1s$  state:  $E_{1s}$

- When two hydrogen atoms, A and B say, approach each other, the  $1s$  wave function  $\Psi_{1s}$  of each individual H atom interact producing two new wave functions,  $\Psi_a$  and  $\Psi_b$  with energies  $E_a$  and  $E_b$  respectively. See Fig.3.1.



**Fig. 3.1**

- The wave functions  $\Psi_a$  and  $\Psi_b$  are obtained by linear combination of the individual atomic orbitals/wave functions (LCAO)

$$\Psi_a = \Psi_{1s}(A) + \Psi_{1s}(B) \quad (3.1)$$

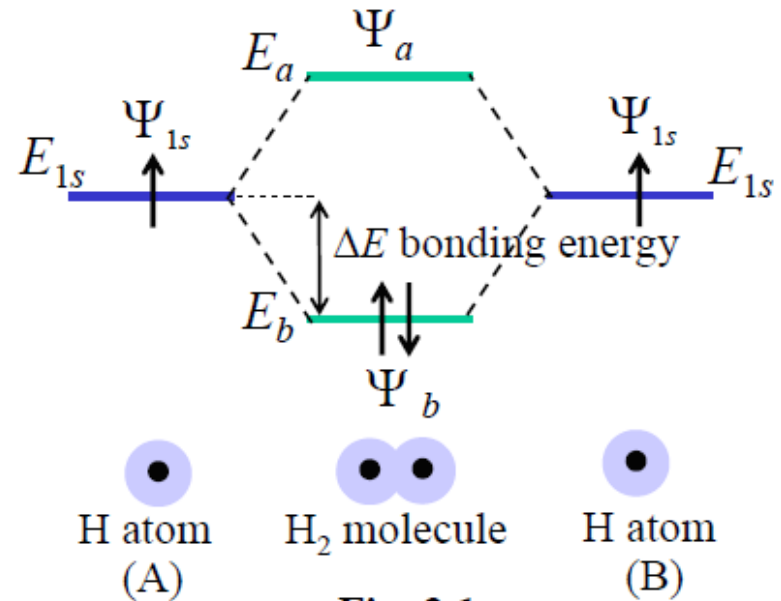
$$\Psi_b = \Psi_{1s}(A) - \Psi_{1s}(B)$$

$\Psi_{1s}(A)$  and  $\Psi_{1s}(B)$  are the  $1s$  wave function of H atoms A and B respectively.

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- The formation of the two new wave functions  $\Psi_a$  and  $\Psi_b$  is consistent with the Pauli exclusion principle, that is: in a given electrons system, there are no two electrons may have the same quantum state.
  - In our case here, this means we cannot have two identical  $\Psi_{1s}$  wave functions in the  $H_2$  molecule.
- The wave functions  $\Psi_a$  and  $\Psi_b$  are called the molecular orbital. Specifically,
  - $\Psi_a$  with energy  $E_a$  is called the anti-bonding orbital
  - $\Psi_b$  with energy  $E_b$  is called the bonding orbital
- Taking the spin into consideration, each wave function will have two distinct states: one spin up and one spin down. In other words, each wave function can accommodate two electrons, one electron with spin up and one electron with spin down.

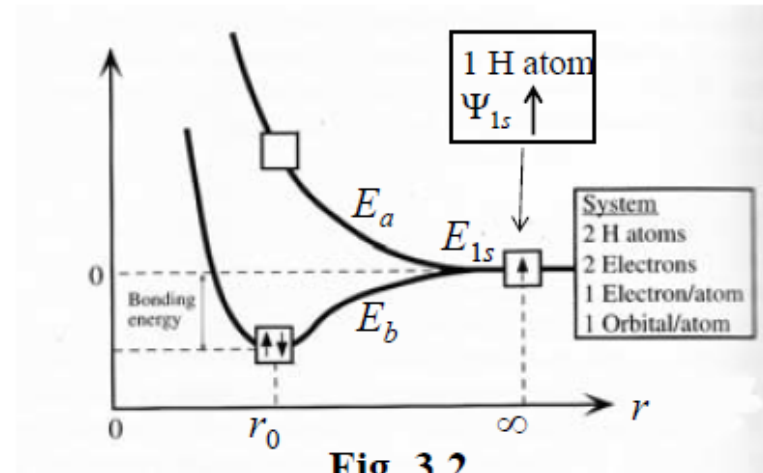
- Before the interaction, each individual H atom has one electron in the  $1s$  state/wave function. After the interaction, in the ground state, the two electrons will occupy the lowest energy wave function  $\Psi_b$ , one electron with spin up and the other electron with spin down.



**Fig. 3.1**

- The lowering of the original energy before the interaction ( $E_{1s}$ ) and the ground state energy after the interaction ( $E_b$ ) is the bonding energy of the H<sub>2</sub> molecule ( $\Delta E$ ).
- The formation of the bonding and anti-bonding energy,  $E_b$  and  $E_a$  respectively, from the original energy  $E_{1s}$  is referred to as energy splitting

- The energy splitting, in other words the separation of  $E_a$  and  $E_b$ , varies with the inter-atomic distance  $r$  between the two H atoms as shown in Fig. 3.2

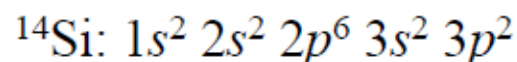


- It is important to note here that:
  - the number of quantum states before interaction = the number of quantum states after interaction
  - the number of electrons before interaction = the number of electrons after interaction

### 3.2 Energy band: conduction and valence bands

Having understood the formation of the molecular orbitals and energies, we can now look into the energy band formation in semiconductor crystals.

- Consider a silicon crystal consisting of  $N$  silicon atoms. The electronic configuration of an isolated silicon atom is

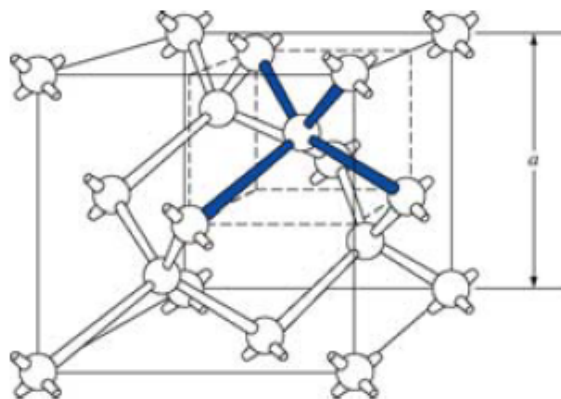


- The number of states and electrons of  $N$  isolated Si atoms is given in Table 3.1

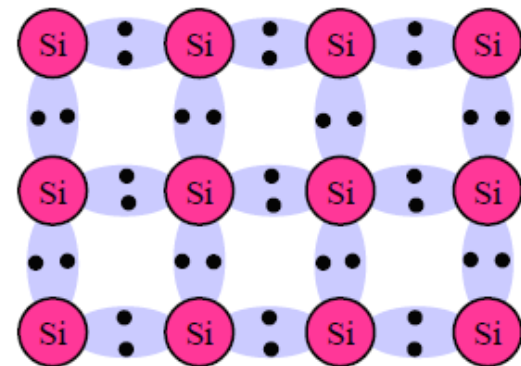
**Table 3.1**

| States | No. of states | No. of electrons |
|--------|---------------|------------------|
| $1s$   | $2N$          | $2N$             |
| $2s$   | $2N$          | $2N$             |
| $2p$   | $6N$          | $6N$             |
| $3s$   | $2N$          | $2N$             |
| $3p$   | $6N$          | $2N$             |

- The  $N$  silicon atoms are now brought together to form silicon crystal. How are these silicon atoms bonded ?
  - They are bonded by the covalent bonding. Each Si atom share its four valence electrons (2 electrons from each  $3s$  and  $3p$  states) with the four nearest neighbors to form stable configuration or to complete their subshells. As a result each Si atom is surrounded by 8 electrons. This is shown in Fig. 3.3



(a) 3-dimensional representation of the covalent bonding in Si crystal at 0 K



(b) 2-dimensional representation of the covalent bonding in Si crystal at 0 K

**Fig. 3.3** Covalent bonding of silicon crystal

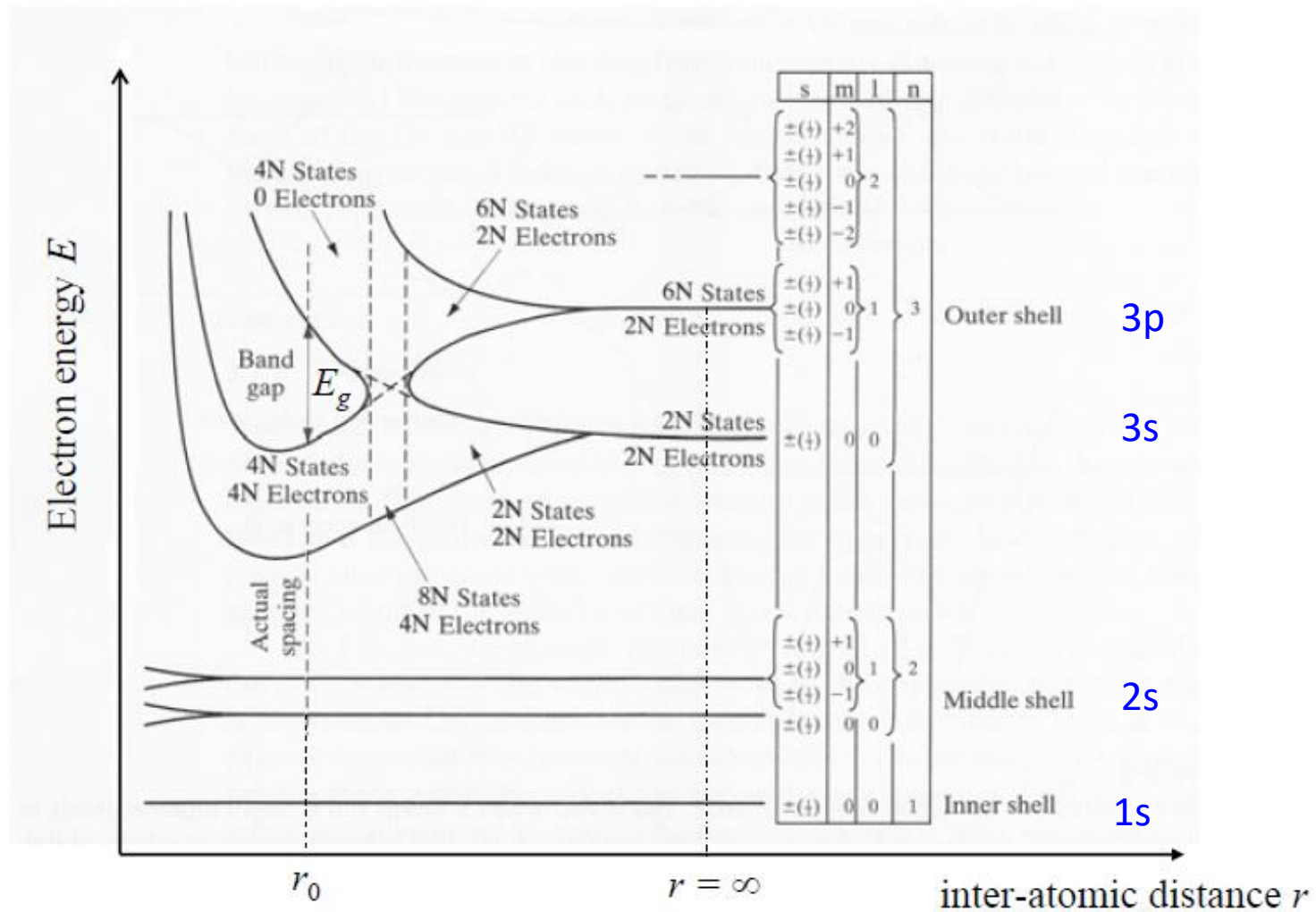


- The formation of the energy band in Si crystal can be explained as follow. As the inter-atomic distance decreases the outer shells  $3s$  and  $3p$  states interact. See Fig.3.4
  - The  $2N$  states of  $3s$  interact. The energy levels split forming  $3s$  band. There are  $2N$  states and  $2N$  electrons in the  $3s$  band.
  - The  $6N$  states of  $3p$  interact. The energy levels split forming  $3p$  band. There are  $6N$  states and  $2N$  electrons in this band.
  - The degree of the energy splitting increases with decreasing inter-atomic spacing
  - The band formation is consistent with the Pauli exclusion principle: no two electrons in a given system may have the same quantum states.



- As the inter-atomic spacing decreases further, the  $3s$  and  $3p$  bands grow and merge into a single band composed of mixture of energy levels. This  $3s$ - $3p$  band contains  $8N$  states and  $4N$  electrons.
- Further decreasing the inter-atomic distance, approaching the equilibrium inter-atomic spacing  $r_0$ , the  $3s$ - $3p$  band splits into two bands separated by an energy gap or a bandgap  $E_g$ .
  - The upper band, called the conduction band, contains  $4N$  states
  - and the lower band, called the valence band, also contains  $4N$  states.
- In a perfect crystal, there are no allowed energy levels (quantum states) in the bandgap for electrons to occupy. The bandgap is also called a “forbidden band”.

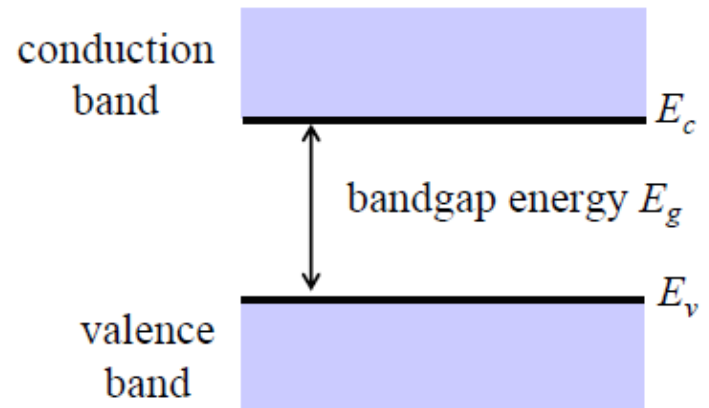
- At 0 K, the electrons will occupy the lowest energy states and therefore all  $4N$  available electrons occupy all  $4N$  states in the valence band. In other words, at 0 K the valence band is fully occupied and the conduction band is empty. In this state, the semiconductor is an insulator, no electrical conduction can take place.
  - for the electrical conduction to take place, electrons must be able to move into empty states. It follows that for the electrical conduction, the electrons must be excited to the conduction band.



**Fig. 3.4**  $r_0$  is the equilibrium inter-atomic spacing

- The energy of the conduction and valence bands versus the physical distance  $x$  of the semiconductor sample can be represented by an energy band diagram as shown in Fig. 3.5

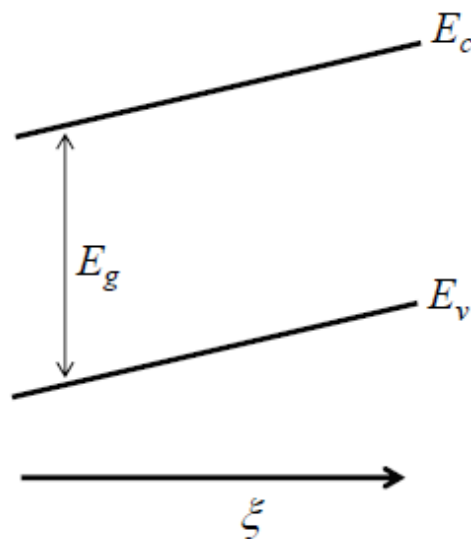
- the energy refers to the electron energy
- the physical distance  $x$  is normally not indicated



**Fig. 3.5**

- $E_c$  is the energy of the bottom of the conduction band, very often called the conduction band edge
- $E_v$  is the energy of the top of the valence band, very often called the valence band edge

- If an electric field  $\xi$  is applied to the semiconductor sample
  - the energy band is not flat anymore as in Fig. 3.5
  - the energy bands will be inclined at a gradient proportional to the electric field according to equation (3.2). The energy band diagram is shown in Fig. 3.6



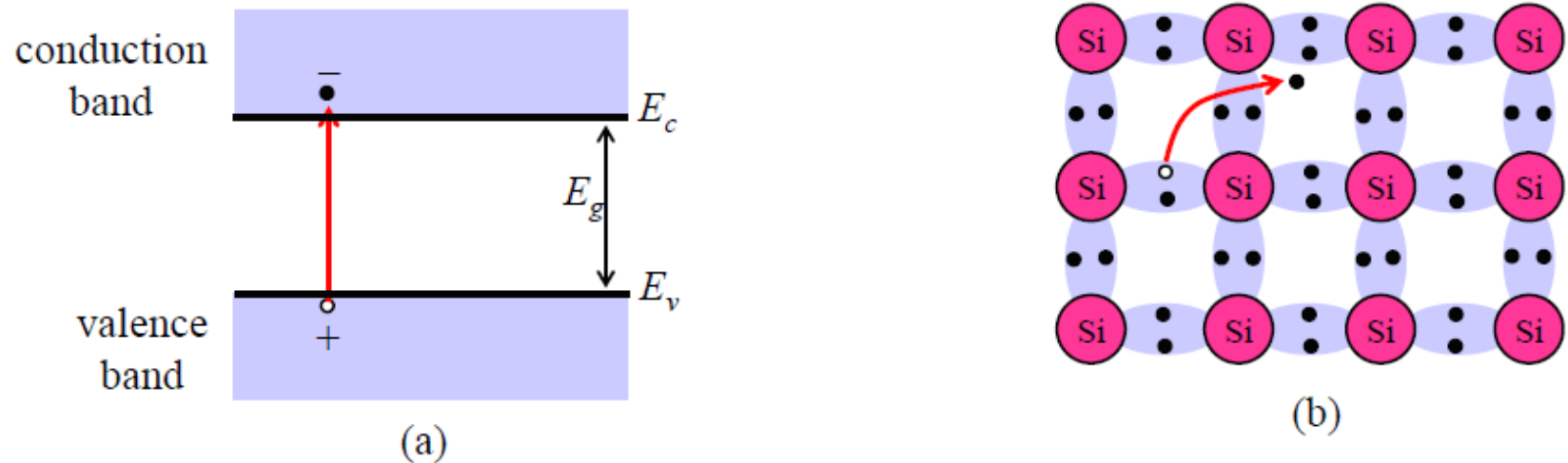
$$\xi = \frac{1}{q} \frac{dE}{dx} \quad (3.2)$$

### 3.3 Electrons and holes

- At  $T = 0$  K, the valence band is fully occupied by electrons and the conduction band is empty. The semiconductor behaves like an insulator.
- The electrons in the valence band can be excited to the conduction by
  - increasing the temperature
  - illuminating crystal with light of a suitable wavelength



- Let us, first of all, consider the excitation of the electrons from the valence band to the conduction by increasing the temperature. As the temperature of the semiconductor is raised above 0 K, i.e.  $T > 0$  K
  - some electrons in the valence band gain enough thermal energy to be excited across the bandgap to the conduction band. This process corresponds to the electrons breaking the Si-Si covalent bonds leaving behind broken bonds. This is illustrated in Fig. 3.7

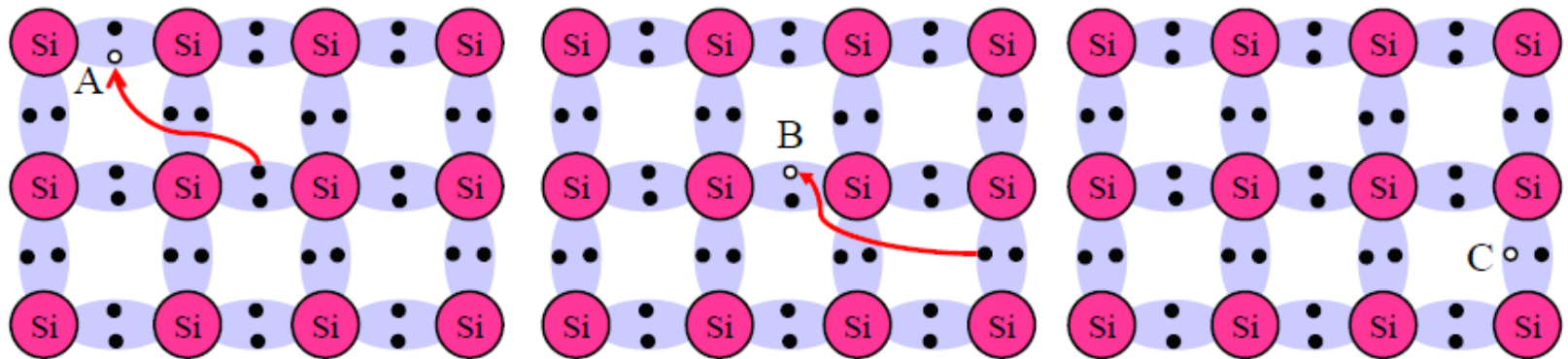


**Fig. 3.7**

- Electron, negatively charged ( $-1.6 \times 10^{-19}$  C)
- hole (broken bond), positively charged ( $+1.6 \times 10^{-19}$  C)

- Some of the states in the conduction band is now occupied by electrons and some of the states in the valence band is now empty. An empty state in the valences band is referred to as a **hole**, it corresponds to the broken bond in the Si-Si bonding. The hole is positively charged with a value of  $1.6 \times 10^{-19}$  C.
- The electrons and holes can “freely” move throughout the crystal. The term free is obviously not in the real sense since the movement of the electrons and holes in the crystal is under the influence of electric fields formed by the ions and electrons in the crystal (compare to the real free electron in vacuum).
- The electrons and holes can only move freely for a short period of time, after which they will collide with vibrating atoms, impurity atoms and defects.

- The movement of the hole is actually the movement of the electron from the Si-Si bond to the broken bond. This is illustrated in Fig. 3.8. It is easier and convenient to think in term of the displacement of holes rather than the actual displacement of electrons from Si-Si bonds to the broken bonds.



**Fig. 3.8** The hole has moved from A to B to C

- The electrons and holes are created by the excitation of electrons from the valence band to the conduction band. This direct excitation of electrons from the valence band to the conduction band is referred to as **direct band-to-band generation**. Obviously, the number of hole in the valence band = the number of electrons in the conduction band. Since the electrons and holes are created in pair, they are also called electron-hole pairs.
- There are empty states in the conduction and valence bands, therefore the electrons and holes are able to perform a directed motion under the action of an external electric field, conducting an electric current.

- The electrons in the valence band can also be excited to the conduction band, creating electron-hole pairs, by illuminating the semiconductor with light having a photon energy  $E_{ph}$  equal or greater than the bandgap energy  $E_g$ . This is illustrated in Fig. 3.9.

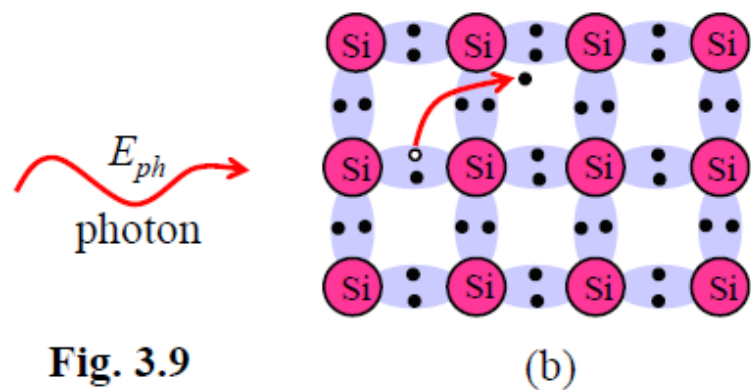
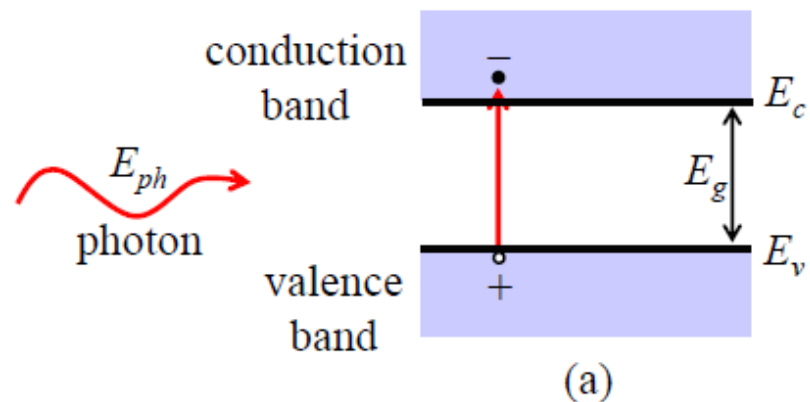
$$E_{ph} \geq E_g \quad (3.3)$$

$$E_{ph} = h\nu = \frac{hc}{\lambda} \quad (3.4)$$

$$\frac{hc}{\lambda} \geq E_g$$

$$\frac{1240}{\lambda} \geq E_g \quad (3.5)$$

where  $\lambda$  in nm and  $E_g$  in eV



**Fig. 3.9**

### 3.3.1 Direct band-to-band generation and recombination

We have learnt that electrons can be excited from the valence band to the conduction band, creating electron-hole pairs, by thermal excitation and illumination (excitation by light).

- What happens if a semiconductor is subjected to a certain temperature  $T$ , or illuminated with light ( $E_{ph} \geq E_g$ ) or both

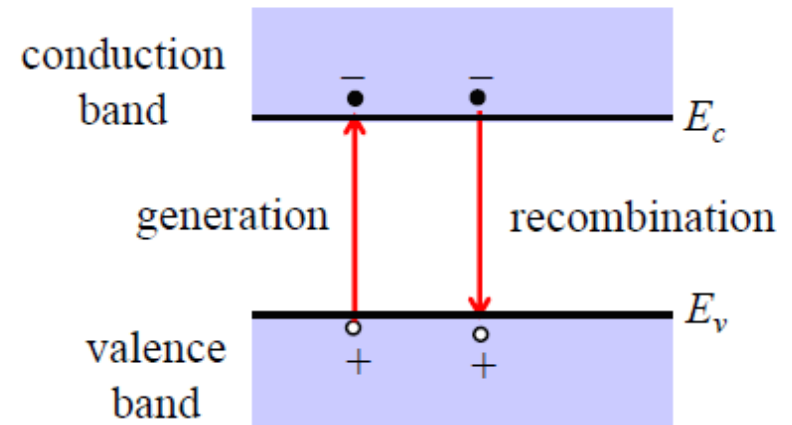
We have to ask ourself a question:

- Do these generated electron-hole pairs keep increasing continuously ?
- The answer is certainly “NO”, otherwise we will observe that all the covalent bonds of the semiconductor will be broken simply by putting a sample in a room. We can subject a semiconductor to room temperature by simply putting the sample in a room.



So, what happens ?

- The reason is that as electron-hole pairs are generated, they also can recombine. In this process the electrons and holes are annihilated (i.e. disappear). The process is called recombination. This is illustrated in Fig. 3.10.
- The direct generation and recombination processes from band-to-band shown in Fig. 3.10 is called direct band-to-band generation and recombination



**Fig. 3.10**



- A steady state will be established when

$$\text{generation rate } G = \text{recombination rate } R \quad (3.6)$$

- The generation rate  $G$  is the number of electron-hole pairs created per unit volume per second (units:  $\#/cm^3\cdot s$ ). The generation is the sum of the thermal generation rate and the generation rate by the external excitation (if any)
- The recombination rate  $R$  is the number of electron-hole pairs recombining per unit volume per second (units:  $\#/cm^3\cdot s$ ).

It is expected that the recombination rate is proportional to the electron concentration in the conduction band and the hole concentration in the valence band.

- Obviously, in the steady state, the electron and hole concentrations are constant with time.

- When a semiconductor is at temperature  $T$  and there is no external disturbance/excitation (e.g. external electric field, magnetic field or light illumination) the electrons and holes are solely created by thermal energy. An equilibrium will be established, and this equilibrium state is called **thermal equilibrium** since there is only thermal energy involved.

In thermal equilibrium:

- thermal generation rate  $G_{th}$  = recombination rate  $R$   
(the generation rate is called thermal generation rate)
  - there exist thermal equilibrium electron and hole concentrations,  $n_0$  and  $p_0$  respectively.
- The thermal generation rate, the recombination rate and the generation rate by the external excitation play a very important role in semiconductors. In chapter 7, we will express the generation and recombination rates mathematically.

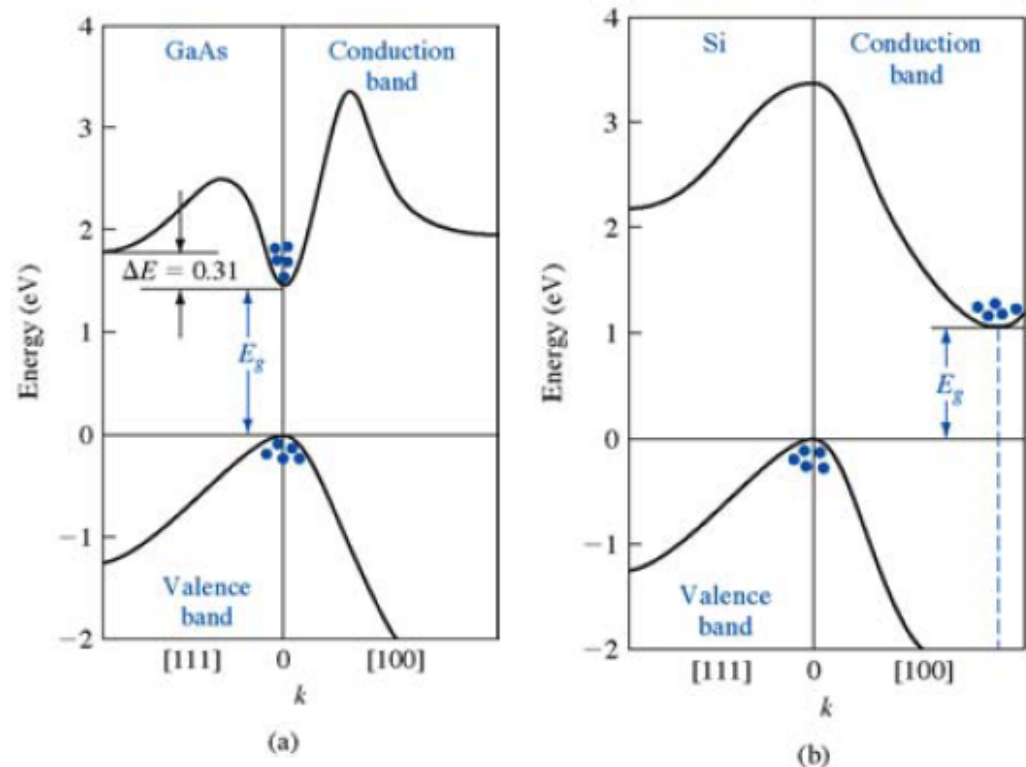
### 3.4 Direct and Indirect Bandgap Semiconductors

Besides the energy band diagram in Fig. 3.5, the electron energy  $E$  can also be related to its momentum  $p$ . This relation will lead to direct and indirect bandgap semiconductors.

### 3.4.3 $E$ versus $k$ ( $E - k$ ) for Si and GaAs

Consider the energy  $E$  versus  $k$  ( $E - k$ ) for silicon and gallium arsenide (GaAs) semiconductors

- $E - k$  is symmetric in  $k$  and that no new information is obtained by displaying the negative axis.
- It is normal practice to plot  $E - k$  along positive  $[100]$  direction to the right and along positive  $[111]$  direction to the left.



**Fig.3.16** Energy band structure of (a) GaAs and (b) Si

Fig. 3.16 shows  $E - k$  plots for GaAs and Si semiconductors

- In the  $E - k$  diagrams we see that
  - the majority of electrons occupy the bottom of the conduction band
  - and the majority of holes occupy the top of the valence band
  - the bandgap energy  $E_g$  is the energy difference between the conduction band minimum and the valence band maximum

- The  $E - k$  diagram for GaAs
  - Conduction band minimum and valence band maximum both occur at same  $k$ -value (in this case at  $k = 0$ ). A semiconductor with this property is called a direct bandgap semiconductor.
  - The electron transitions from conduction band to valence band or vice versa require the conservation of energy and momentum.
  - In the direct bandgap semiconductor, the transitions do not require changes in the crystal momentum. Therefore, the transitions are very efficient. For the transitions from the conduction band to the valence band, the energy is given off as photon (light).
  - Consequently, the direct band gap semiconductors are suitable for light producing optical devices (e.g. semiconductor laser)

- The  $E - k$  diagram for Si
  - Conduction band minimum and valence band maximum do not occur at same  $k$ -value. A semiconductor with this property is called an indirect bandgap semiconductor.
  - The electron transitions from the conduction band to the valence band require changes in the crystal momentum.
  - The crystal momentum change can be satisfied, in principle, through the involvement of phonons (lattice vibration) in the transition process. Because of the need of the phonons involvement, this type of transitions are not efficient.
  - In the presence of the defect states, the transitions of electrons from the conduction to the valence band are more likely through defect states and the energy is given out as heat.
  - This is why the indirect semiconductor materials are not suitable for light producing devices.